

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: Fundamentals of Data Science **COURSE CODE:** DSA04

COURSE OUTCOME COVERED

CO1: Analyse the role of data science, facets of data, and the big data ecosystem for informed decision-making. (BL4)

QUESTIONS: 05

Total Marks:100

Annexure A

CONCEPT MAPPING

Code of Conduct:

I G. Sathya (192472133) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Concept Mapping

1. A college wants to analyze student performance using attendance records, internal marks, assignment scores, and final exam results. Draw a concept map showing the Data Science process from data collection to data transformation.
2. A hospital collects patient details from registration forms, lab reports, medical devices, and doctor notes. Construct a concept map showing the different data collection sources and how they are used for analysis.
3. An online shopping company finds missing customer details, duplicate orders, wrong product entries, and unusual purchase amounts in its dataset. Draw a concept map showing the data cleaning steps required before analysis.
4. A smart city project collects traffic data from CCTV cameras, GPS devices, sensors, and mobile applications. Prepare a concept map showing how data from multiple sources can be integrated into a single dataset.
5. A bank wants to prepare customer data for loan approval prediction. The dataset contains income, age, job type, loan amount, and credit score. Draw a concept map showing suitable data transformation techniques such as encoding, normalization, and feature scaling.

RUBRICS FOR CALCULATION

Concept Mapping					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

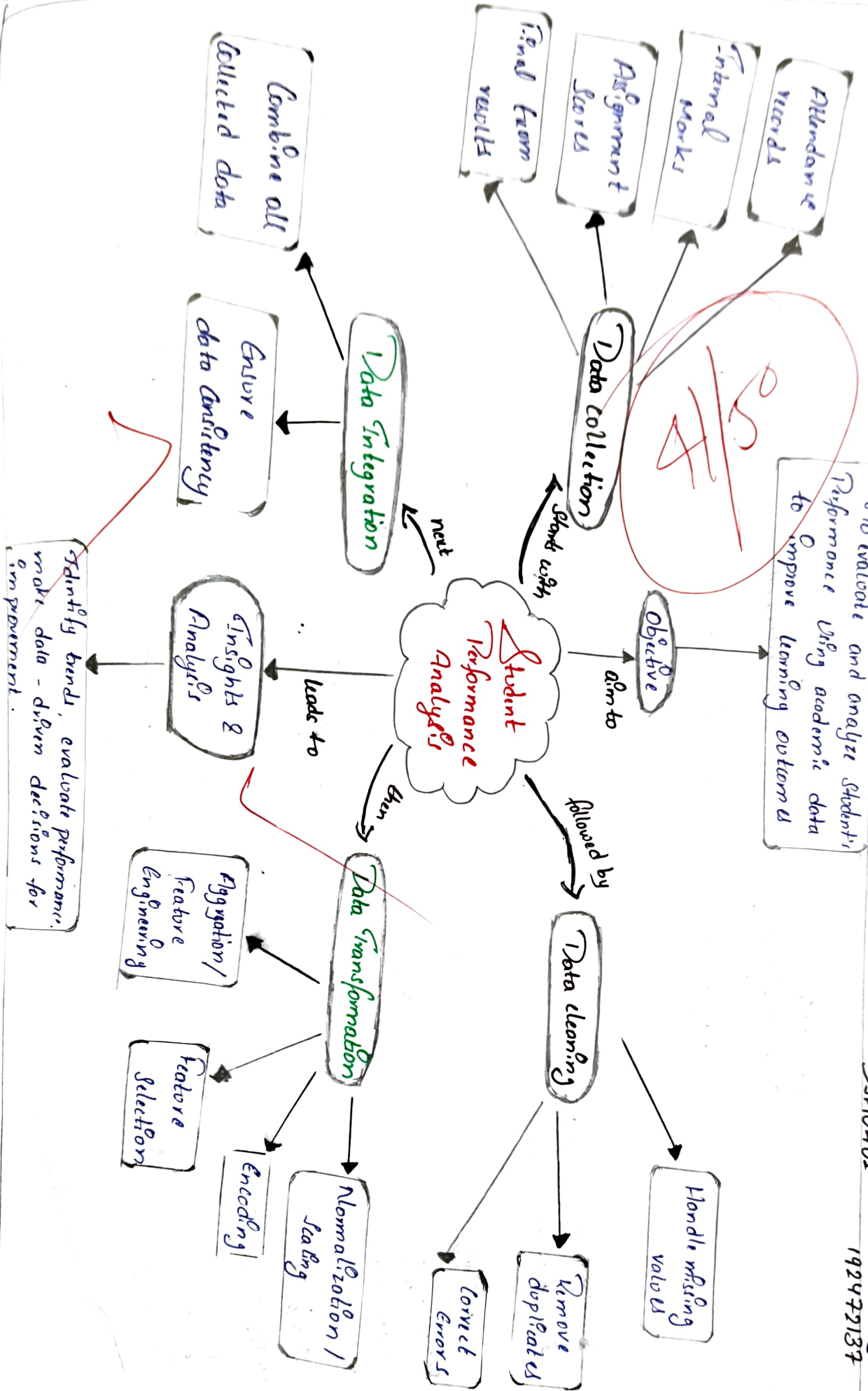
CO1-ASSESSMENT TOOL -2 Concept Map

QUESTIONS : 5

MARKS : 50

Q No	Concept Identification (25%)	Linkages (25%)	Organisation (20%)	Integration of Literature (20%)	Clarity (10%)	Total Score (100%)
1	2.5 /2.5	2.5 /2.5	1 /2.0	1 /2.0	1 /1.0	7 /10
2	2.5 /2.5	2.5 /2.5	1 /2.0	1 /2.0	1 /1.0	7 /10
3	2.5 /2.5	2.5 /2.5	1 /2.0	1 /2.0	1 /1.0	8 /10
4	2.5 /2.5	2.5 /2.5	1 /2.0	1 /2.0	1 /1.0	9 /10
5	2.5 /2.5	2.5 /2.5	1 /2.0	2 /2.0	1 /1.0	10 /10
Overall Rubrics Value	12.5 /12.5	12.5 /12.5	5 /10	6 /10	5 /10	41 /50
	25 /25	25 /25	10 /20	12 /20	10 /10	82 /100

To evaluate and analyze student's performance using academic data to improve learning outcomes



1. Registration form

2. IHB reports

3. Medical Review

4. Doctor Notes

Data Collection Sources

Obtains data from

Purpose
to collect, organize and analyze patient data to improve healthcare services and decision making

Hospital Data Collection & Analysis

for

How Used for Analysis

• Patient care improvement

• Disease Pattern Identification

• Treatment Effectiveness Evaluation

• Report Generation & decision making

Data Handling Processes

Supported by

Data Storage

(Secure database)

Data Integration

(Merging from multiple sources)

Data Validation

(Checking Accuracy & completeness)

Q3 Data Cleaning Steps in Online Shopping Dataset

Raw E-commerce Dataset

- * Customer
- * orders
- * Product
- * Transaction

Missing Customer

Filling missing value (Using mean / median) or remove records

Duplicate orders

Identify duplicate (Using order ID) and remove duplicate records

Wrong Product Entries

Correct product name category, price and both wrong entries

Unusual purchase Amount

Detect outliers Using IQR Using z-score

Ready Analysis

Dataset of cleaned

Data validation

Key takeaway clean data improve accuracy reliability & performance.

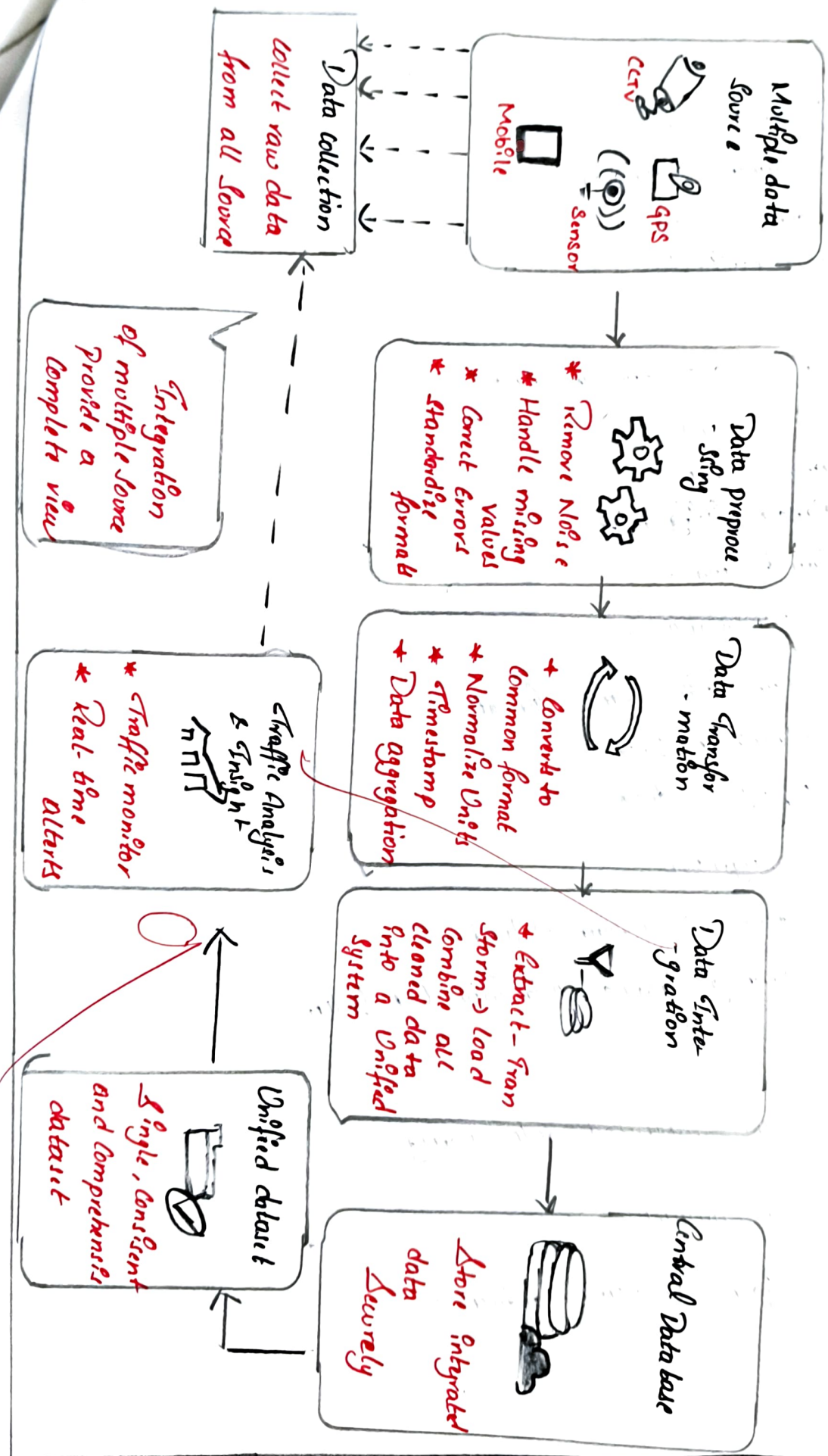
High equality reliable data for analysis

- * Consistency check
- * Accuracy check
- * Format check

Steps Involved:

1. Raw e-commerce dataset.
 - * Contains customer details, order, products and transaction.
2. Handling Missing customer detail
 - * Fill missing values using mean / median / mode.
 - * Or remove ~~dup~~ incomplete records.
3. Removing duplicate orders.
 - * Identify duplicates using Order ID.
 - * Remove repeated entries to avoid incorrect analysis.
4. Correction Wrong product entries
 - * Fix incorrect product names, categories or prices.
 - * Ensure consistency in product data.
5. Handling Unusual purchase amount.
 - * Detect using Z-Score or IQR.
 - * Treat by capping transforming or removing.
6. Data validation
 - * Consistency check * Accuracy check * Format check.
7. Cleaned dataset
 - * High-quality, reliable data
8. Ready for analysis
 - * Used for decision-making and business insight.

Q4 Integrating Multiple Data Source in Smart City Project



Steps Involved

1. Multiple data Source

- + CCTV camera (video data)
- * GPS device (location)
- + Mobile application

2. Data collection

- + Gather raw data from all source

3. Data preprocessing

- * Remove noise
- * Handle missing value
- * Correct errors.

4. Data transformation

- + Convert into common format
- + Normalize units
- * Aggregate data

5. Data integration

- * Extra → Transform → Load
- * Combining cleaned data into One system.

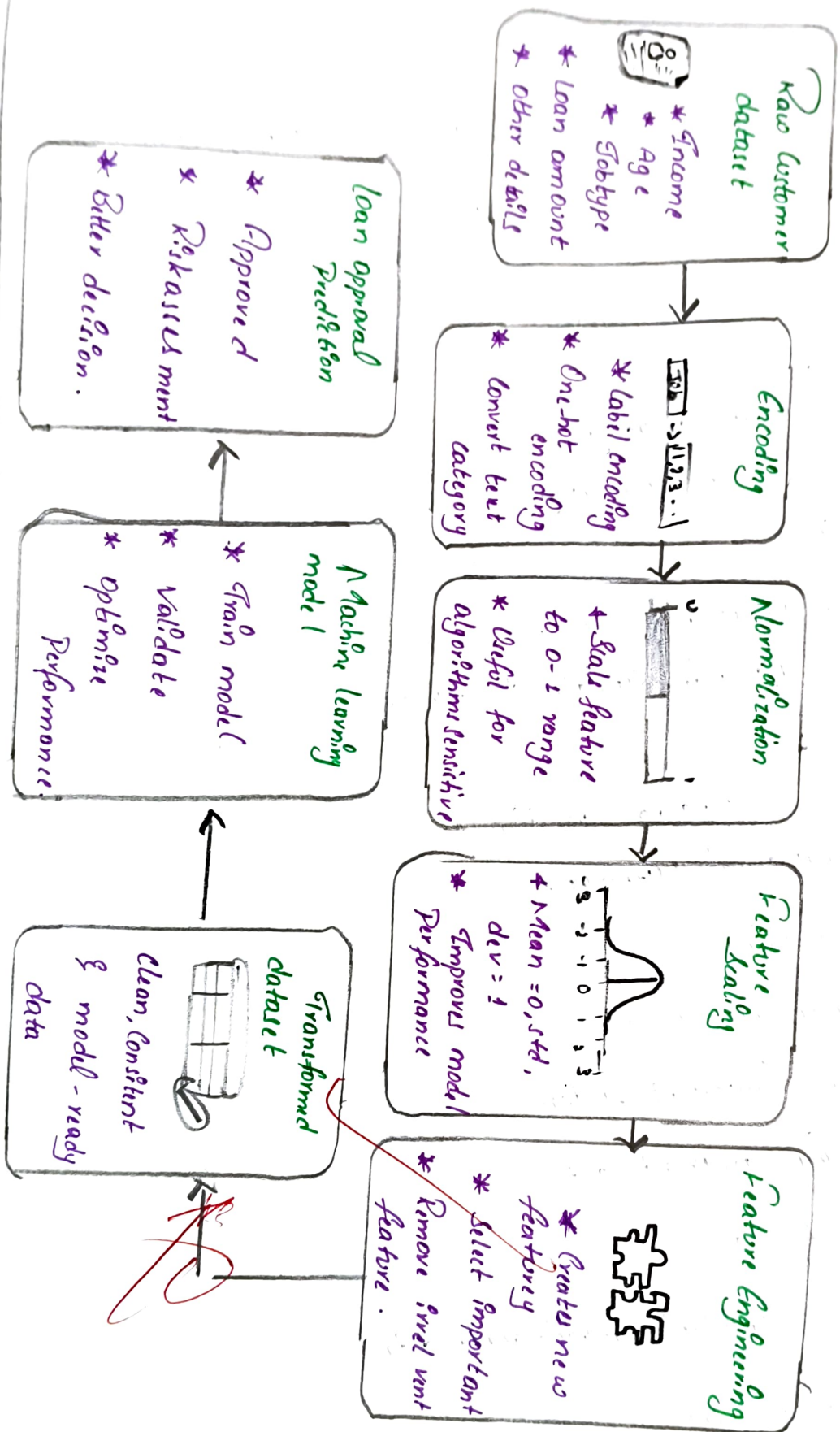
6. Central Database

- * Store integrated data securely

7. Unified dataset

- + Single, Consistent dataset.

Q5 Data Transformation for Loan Approval Prediction



Steps Involved

Data transformation prepare raw data for machine learning model.

Encoding:

* Convert job type into numbers

* Methods:

→ label encoding

→ One-hot encoding

Normalization

* Scales values b/w 0 & 1.

* Helps algorithm sensitive to scale.

Feature Scaling

* Mean = 0, Standard deviation = 1

* Improves model performance.

Feature Engineering:

* Create new features

* Select important feature

* Remove irrelevant data

Transformation data

* Clean, Consistent, model-ready data

Loan Approval prediction

* Output → Approved, risk assessment better decision making.